

A finite noise window for concentration of a shared flow

Exact stationarity and the finite-population correction

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Abstract

State-dependent diffusion changes how often a channel occupies a given state. When channels share a fixed total throughput, their population distribution and their flow distribution can differ substantially. We study a bounded, reflected Itô model with independent noise, share proportional to g_i^m , and diffusivity $D_i = Ng_i^q / \sum_j g_j^m$. Its exact joint stationary density is proportional to $(\sum_j g_j^m) \prod_i g_i^{-q}$. The tagged-channel density is $p_N = (1 - 1/N)p_0 + f/N$, where $p_0 \propto g^{-q}$ and $f \propto g^{m-q}$; f is also the pooled flow density. The population fraction above the pooled half-flow threshold tends to zero as both state range and population grow if and only if $1 \leq q \leq m + 1$, for exact powers. At fixed population it instead approaches $1/(2N)$ inside the window. The upper boundary is the familiar weighted-moment threshold, here given an exact conserved-flow interpretation. It limits a specific noise-driven concentration mechanism; it is neither a universal noise limit nor a result about persistent winners. Independent event simulations check the stationary predictions on bounded grids. Empirical applicability remains untested.

1 What the upper boundary says

A collection of paths can be counted in two ways: by the number of paths in each state, or by the fraction of throughput they carry. A rare large state can matter little to the first count and greatly to the second. The question here is when that difference becomes extreme as the accessible state range grows.

Noise affects this comparison through residence times. In a scalar driftless Itô diffusion with reflecting boundaries, the stationary density is inversely proportional to diffusivity. State-dependent noise and occupation probabilities have a long history [1, 2]. Weighting a power-law distribution by a power of its argument introduces a standard moment threshold [3]. Neither ingredient is new.

Newman explicitly discusses both the general moment condition and the fraction of a weighted total held in a population tail (Secs. III.B and III.D of Ref. [3]). Their use for a fixed shared flow requires care: each channel's share depends on every other channel, and the same normalization can change the physical clock of the state dynamics. We give an exactly solvable coupled model in which this dependence is explicit. The result identifies the finite-population correction and makes the observable behind the noise window precise.

For $m = 1$, the upper edge is $q = 2$. Beyond it, increasing the upper state cutoff does not make a vanishing population fraction account for half the pooled flow. This does not mean that flow becomes equally distributed. A small but nonzero fraction can still carry half. Nor is q a noise amplitude: it describes how diffusivity changes with state. Multiplying every diffusivity by one constant changes relaxation times, but leaves all stationary results below unchanged.

2 A model with conservation imposed at every instant

Let $N \geq 1$ channels have dimensionless states $g_i \in [1, R]$, where $R > 1$. The channels are supplied in advance. No spatial energy-gradient field, channel creation or channel extinction is modeled. The calculation therefore does not establish spontaneous channel formation in a driven material. For $m > 0$, define the total weight S and the actual flow through channel i by

$$S(\mathbf{g}) = \sum_{j=1}^N g_j^m, \quad J_i(\mathbf{g}) = I \frac{g_i^m}{S(\mathbf{g})}. \quad (1)$$

Thus $\sum_i J_i = I$ for every configuration, not only in expectation. We choose the interior dynamics

$$dg_i = \sqrt{2D_i(\mathbf{g})} dW_i, \quad D_i(\mathbf{g}) = \frac{N g_i^q}{S(\mathbf{g})}, \quad (2)$$

with independent Wiener processes, normal reflection at both endpoints, and zero interior drift. Boundary local-time terms are implicit. A positive common prefactor has been absorbed into the time unit. The lower cutoff sets the state unit; all powers refer to this specified coordinate.

The normalization in Eq. (2) is a constitutive assumption. On the line $q = m$, $D_i = (N/I)J_i$, so noise variance rate is proportional to actual flow. Away from that line the model keeps the common factor $1/S$. The apparently similar choice $D_i \propto J_i^{q/m}$ instead contains $S^{-q/m}$ and defines different dynamics. Conservation of flow alone does not select either rule.

The Itô convention would follow from an appropriate small-kick limit of conditionally unbiased additive events in g , evaluated at the pre-event state. Such a limit also needs control of temporal correlations and higher moments. Unresolved deterministic dynamics do not automatically satisfy these conditions. Under a nonlinear change $y = h(g)$, Itô's formula adds the drift $D(g)h''(g)$ even when the drift in g is zero. The zero-drift assumption must therefore be justified in the measured coordinate, rather than restored after arbitrary relabelling.

3 Exact stationary population and flow distributions

On $[1, R]$, write

$$Z_R = \int_1^R g^{-q} dg, \quad p_0(g) = \frac{g^{-q}}{Z_R}, \quad \mu_R = \int_1^R g^m p_0(g) dg, \quad f_R(g) = \frac{g^m p_0(g)}{\mu_R}. \quad (3)$$

All quantities are finite at finite R , for every real q .

Proposition 1. *The unique stationary density of Eq. (2) on the reflecting box is*

$$\pi_{N,R}(\mathbf{g}) = \frac{S(\mathbf{g})}{N\mu_R} \prod_{j=1}^N p_0(g_j). \quad (4)$$

Its tagged-channel marginal and its pooled flow density are, respectively,

$$p_{N,R}(g) = \left(1 - \frac{1}{N}\right) p_0(g) + \frac{1}{N} f_R(g), \quad (5)$$

$$\mathbb{E}_\pi \left[\sum_{i=1}^N \frac{J_i}{I} \delta(g - g_i) \right] = f_R(g). \quad (6)$$

Proof. Integration of $S \prod_j p_0(g_j)$ gives $N\mu_R$, establishing normalization. For each coordinate,

$$D_i \pi_{N,R} = \frac{1}{Z_R \mu_R} \prod_{j \neq i} p_0(g_j)$$

is independent of g_i . Every probability current $-\partial_i(D_i\pi_{N,R})$ therefore vanishes, including at the boundaries. The diffusion is uniformly nondegenerate on the connected compact box, giving uniqueness. Expanding the sum S and integrating out the other coordinates gives Eq. (5). Multiplication by $J_i/I = g_i^m/S$ cancels S in the joint law; integration and summation give Eq. (6). $\square$

The calculation also has a simple clock interpretation. If $L_0 = \sum_i g_i^q \partial_i^2$ is the generator of independent reflected channels, the coupled generator is $L = (N/S)L_0$. Configurations with larger S are traversed more slowly relative to that independent process; their stationary weight acquires the factor S . This explains both the solvability and the dependence of the result on the chosen normalization.

An exact independent sampling construction is useful: choose one channel uniformly, draw its state from f_R , and draw the other $N - 1$ states independently from p_0 . Averaging over the chosen channel reproduces Eq. (4). That designation is resampled for each configuration; it does not identify a permanent winner.

The correction matters even in simple cases. For $N = 1$ and $q = m$, the channel carries all the flow and has constant diffusivity. Its stationary density is uniform, not g^{-m} . At fixed R , the tagged marginal approaches p_0 as $N \rightarrow \infty$. The independent-channel calculation is recovered in that limit.

4 The concentration observable and its window

Let a_R be the state threshold above which half of the pooled flow lies:

$$\int_{a_R}^R f_R(g) dg = \frac{1}{2}. \quad (7)$$

Define $P_0(R) = \int_{a_R}^R p_0(g) dg$ and $P_N(R) = \int_{a_R}^R p_{N,R}(g) dg$. Equation (5) gives

$$\boxed{P_N(R) = \left(1 - \frac{1}{N}\right) P_0(R) + \frac{1}{2N}.} \quad (8)$$

This counts a population fraction above one threshold fixed across stationary observations. It is a pooled observable, with each observation weighted equally in physical time.

Theorem 1. *For Eqs. (1)–(2) with exact powers and $m > 0$, $P_N(R) \rightarrow 0$ as both $R \rightarrow \infty$ and $N \rightarrow \infty$ if and only if*

$$\boxed{1 \leq q \leq m + 1.} \quad (9)$$

At fixed N , the limit inside this window is $1/(2N)$.

Proof. Put $A = m - q + 1$ and $B = 1 - q$. Direct integration gives

$$a_R = \begin{cases} [(R^A + 1)/2]^{1/A}, & A \neq 0, \\ \sqrt{R}, & A = 0, \end{cases} \quad P_0(R) = \begin{cases} (R^B - a_R^B)/(R^B - 1), & B \neq 0, \\ \log(R/a_R)/\log R, & B = 0. \end{cases} \quad (10)$$

For $q < 1$, $a_R/R \rightarrow 2^{-1/A}$ and $P_0 \rightarrow 1 - 2^{-B/A} > 0$. At $q = 1$, $P_0 \sim \log 2/(m \log R) \rightarrow 0$. For $1 < q < m + 1$, a_R grows proportionally to R , while the normalized population tail above it vanishes. At $q = m + 1$, $P_0 = 1/(R^{m/2} + 1) \rightarrow 0$. For $q > m + 1$, $a_R \rightarrow 2^{1/(q-m-1)}$, so $P_0 \rightarrow 2^{-(q-1)/(q-m-1)} > 0$. Equation (8) supplies the finite- N result and the joint limit. $\square$

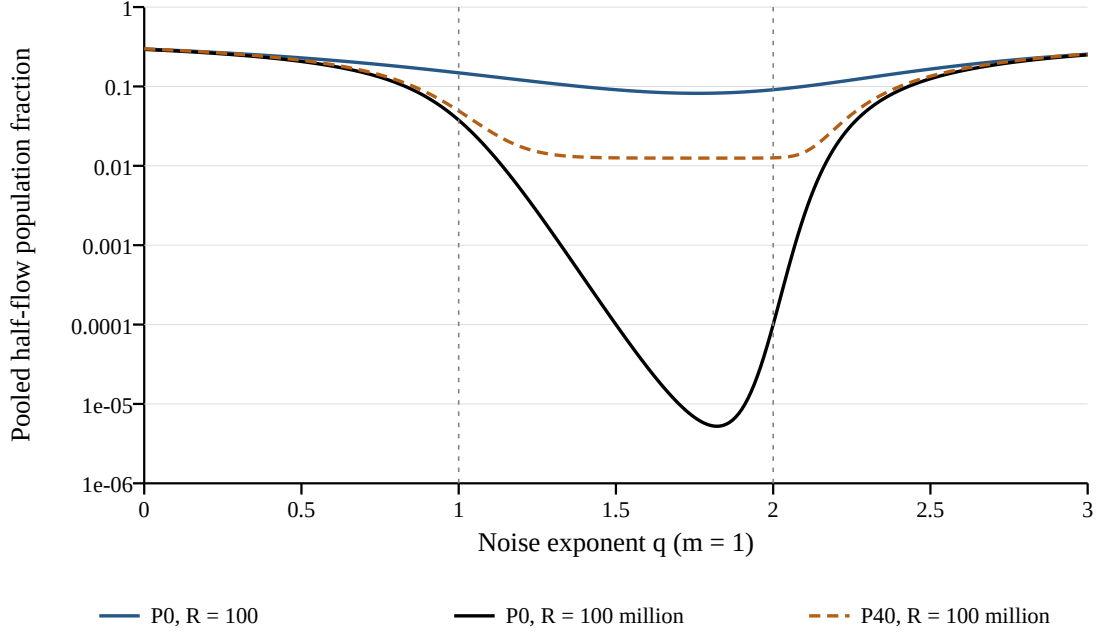


Figure 1: Exact stationary predictions, not empirical measurements or large-range dynamic simulations. The asymptotic window is between the dotted lines. Finite range smooths both boundaries; finite population supplies a floor. P_0 is the large-population limit at each fixed R . The curve at $R = 10^8$ is not the infinite-range limit.

Either iterated limit gives this classification, as does any sequence with both R and N tending to infinity. Stationarity is taken at each bounded pair (N, R) first. No unbounded diffusion or uniform equilibration time is asserted. The same classification holds for any fixed pooled flow fraction strictly between zero and one, with the corresponding fraction divided by N replacing $1/(2N)$.

The integrals behind the two edges are $\int g^{-q} dg$ and $\int g^{m-q} dg$. At the lower edge the population normalization is logarithmic; at the upper edge the flow normalization is logarithmic. The upper-edge population fraction itself decays as a power of R . Figure 1 displays the finite-range expressions.

4.1 Pooled concentration is not a snapshot winner count

In a single configuration, rank shares in descending order and count the smallest number of whole channels whose shares sum to at least one half. Call that count divided by N H_N . It always satisfies $H_N \geq 1/N$. In contrast, P_N can approach $1/(2N)$ because its fixed threshold need not carry half the flow in every snapshot. These observables must not be substituted for one another. The numerical checks measure $\mathbb{E}H_N$ separately against the exact joint sampler. Neither statistic measures how long a particular channel remains dominant. Stationary concentration alone establishes neither persistent structure nor Darwinian selection.

For example, at $m = 1$, $q = 2$, $R = 10^8$, $P_0 \simeq 0.000100$, whereas $P_{40} \simeq 0.012598$. This is a substantial relative correction to the tail fraction, even though $1/N$ is small.

4.2 Limiting slopes do not determine the endpoints

The inclusive endpoints require exact powers or additional tail assumptions. On $g \geq e$, consider the unnormalized flow weight $w(g) = g$ and noise factor $d(g) = g^2(\log g)^2$. Their limiting log-log slopes are $m = 1$ and $q = 2$, but $\int_e^\infty w(g)/d(g) dg$ converges. The upper-edge separation

fails. This example also fits the coupled construction with $S = \sum_i w(g_i)$ and $D_i = Nd(g_i)/S$: replace p_0 by the normalized $1/d$ and f_R by the normalized w/d in the proof. Logarithmic corrections do not change the strict interior power inequalities, but can decide the boundary cases. An estimated slope near an edge is consequently insufficient to classify a measured system.

5 What would make the result physically useful?

The upper bound excludes a proposed mechanism when the measured disturbance scaling is too steep relative to the flow weight, provided the rest of the model applies. It also distinguishes state-dependent disturbance from monotonic reinforcement intuition. At $m = 1$, the large-range, large-population half-flow fraction vanishes for $q = 2$, but approaches $1/8$ for $q = 2.5$. Above the boundary it can still be very small, especially close to $q = 2$. The mathematical transition is the loss of a vanishing fraction, not an abrupt return to equal sharing in a finite apparatus.

The basic inequality is a standard power-law moment condition [3]. Its value here is the connection to actual conserved shares and the observable correction in Eq. (8). These make a conditional prediction that can be checked. They do not establish a new general law of nature, and no priority claim for the elementary threshold is made.

One possible microscopic route to state-dependent variance uses contact patches. If each patch contributes an unbiased additive perturbation in the same measured coordinate, n independent patches give variance proportional to n ; perfectly coherent patches within a channel give variance proportional to n^2 . Inferring $q = m$ or $q = 2m$ additionally requires $n \propto g^m$, fixed coupling per patch, and a suitable time scale. It does not derive the common $1/S$ clock in Eq. (2). In particular, a rule proportional to squared actual flow would have a different normalization. Coherence within a channel must also be distinguished from correlations between channels, which violate the independent-noise assumption.

A fair physical test needs simultaneous channel states and individual flows, with a fixed total and a stable channel population. It must estimate both conditional drift and variance rate in the chosen coordinate, check cross-channel correlations, and allow enough relaxation before comparing stationary statistics. For Eq. (2), conditioning on the configuration or accounting for S is necessary: a regression on g_i alone need not recover q . Drift, measurement noise, jumps, changing populations and limited state range can each prevent the test from addressing this model. Power-law fits themselves also need checks against alternatives [4].

A useful experiment would vary the state dependence of disturbance across the upper edge while measuring the finite-range prediction, rather than expecting an infinite-limit discontinuity. No physical dataset is used as validation in this paper.

6 Numerical checks and their limits

The repository implements a continuous-time nearest-neighbour chain on a uniform state lattice with spacing h . Each allowed jump of channel i by $+h$ or $-h$ has rate

$$\lambda_i(\mathbf{g}) = \frac{Ng_i^q}{S(\mathbf{g})h^2}. \quad (11)$$

Outward boundary jumps are absent. Exponential holding times and rate-weighted event selection simulate this finite chain without a time-step approximation. The lattice invariant law is exactly Eq. (4) with sums replacing integrals. Grid refinement remains necessary for a continuous-state comparison. Observations are taken at fixed physical times, because sampling only jump events would bias residence-time statistics.

N	Grid intervals	Occupancy CDF	Flow CDF	Mean snapshot fraction
8	4	0.017014	0.021730	0.011423
8	8	0.020299	0.024502	0.005979
40	8	0.009496	0.008539	0.001585

Table 1: Largest final absolute errors across the six exponent pairs and four initial conditions for each retained run, all on $[1, 3]$. CDF errors compare with the exact lattice law. The snapshot reference uses 100,000 independent joint-law samples per pair. Seeds are pooled at the final observation time.

Eight automated tests include detailed balance on every edge of an enumerated three-channel, three-state system; independent marginal calculations; the single-channel case; the finite-population quantile identity; sampling checks; continuum integral comparisons; grid refinement; and relaxation from both boundaries in a simple control. The retained ensemble runs use six exponent pairs: $(1, 0.5)$, $(1, 1)$, $(1, 1.5)$, $(1, 2)$, $(1, 2.5)$ and $(2, 4)$. For each pair they use lower-boundary, upper-boundary, log-uniform and exact stationary initial conditions, with three seeds and 128 independent ensembles per seed. Physical observation times are 0, 1, 4, 16.

All retained final errors are below the declared tolerance of 0.05 (Table 1). This is an acceptance criterion, not a confidence interval. Separate exact-law calculations check grid refinement and evaluate large state ranges; they are not dynamical runs at those ranges. The checks support the implementation on the tested bounded grids. They do not establish large-range equilibration times or empirical applicability.

The earlier independent-walker check is excluded from this evidence. It started log-uniformly, which is already the target stationary law at $q = 1$, and did not establish relaxation. It is retained in the repository as a labelled legacy diagnostic. The exact power integrals remain valid.

7 Scope and relation to the wider repository

Scalar diffusion with drift has the zero-current law

$$p(g) \propto \frac{1}{D(g)} \exp\left(\int^g \frac{v(x)}{D(x)} dx\right). \quad (12)$$

For $D = d_0 g^q$ and $v = a g^{q-1}$, this shifts the scalar exponent to $q - a/d_0$. It does not give a general drift correction for the coupled model above. Likewise, scalar accumulation near an upper cutoff does not prove that one channel takes over a shared total. Coupled drift and winner persistence remain separate problems.

Preferential attachment [5] concerns growing networks with attachment rules. Its linear boundary is not a derivation of this model’s lower noise edge. Wealth condensation [6] couples exchange and multiplicative growth; the variable being condensed and its dynamics differ from the fixed-throughput observable used here. These are related comparisons, not equivalent mechanisms.

Beyond the paper, the repository explores inherited states, copying funded by throughput, memory loss and transport of stored states. If throughput supplies usable work at rate ηJ and each copy costs E_{copy} , ideal sustained operation gives a copy rate $\eta J/E_{\text{copy}}$. Inheritance, losses, resets and population replacement must still be specified. Those exploratory models ask how concentration could connect to reproduction. Their conclusions do not follow from the stationary theorem and are not evidence for its physical premises. The README distinguishes the tested paper implementation from these extensions and historical claims.

8 Conclusion

For the specified reflected conserved-flow diffusion, the exact stationary law gives a finite noise window and a finite-population correction. The upper bound states when rare large states cease to dominate pooled throughput in the large-range limit. Its algebra is established moment mathematics. The paper’s contribution is to make that interpretation exact for a coupled model and to provide finite-system predictions and reproducible checks. Whether this model describes a physical system remains an experimental question.

Code and reproducibility. The editable manuscript, compiled PDF, figure data and generation script are in `paper/` at <https://github.com/divadwg/selection-as-physics>. The stationary implementation and tests are in `sims/validation/` and `tests/`; retained data are in `docs/validation_results/`. The underlying validation was committed as `e802c6f`. Build and run instructions are in `paper/README.md`. This revision supersedes the August 2026 draft’s uncorrected finite-population claim, limiting-slope endpoint claim, independent-walker convergence claim and coupled-takeover interpretation.

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